

Experts

“For Precision Timing, MEMS Is The Horse To Bet On Technologically”

SiTime positions itself as a leader in precision timing devices, asserting that their MEMS devices surpass traditional Quartz counterparts. To understand why that matters and to whom, Yashasvini Razdan and Nitisha Dubey spoke to Piyush Sevalia, Executive Vice President of Marketing, SiTime Corporation. Here are key excerpts from their conversation...



PIYUSH SEVALIA
EXECUTIVE VICE PRESIDENT OF MARKETING, SITIME CORP.

How would you explain the category of semiconductors known as ‘precision timing’ to a non-technical CEO of your customer?

In the realm of electronics, precision timing acts as the heartbeat—provided by a timing chip. For example, in a PC with an i7 processor operating at up to 3.9GHz, the frequency of operation, 3.9GHz, is the heartbeat, and all the functions that the processor or the PC perform are referenced to this heartbeat. Every electronic device needs one or more timing chips to operate correctly. Precision timing makes a system run faster, more intelligently, and, in some cases, more safely, as in the case of automated driving. Automated driving won’t be safe until you have a massive infrastructure that is going to allow the car to make decisions. The timing needs to be accurate to make critical decisions in an electronic system quickly.

How is the fast evolution of electronics tied into the solutions you offer?

Our precision timing devices are programmable to accommodate

any frequency within their specified range. For instance, if a device has a range from 1 megahertz to 220 megahertz with no limitations, the device can be programmed to any frequency within this range, with up to 6 decimal places of accuracy. Additionally, voltage levels such as 1.8 volts, 2.5 volts, or 3.3 volts can be programmed, along with various output signalling levels. This level of customisation allows customers to adjust frequency for Ethernet systems to minimise error rates or to customise for non-standard frequencies, regardless of the processor or Ethernet chip they use.

While obtaining a custom frequency from a standard supplier could take 12 to 16 weeks, our approach allows for on-the-spot programming, be it 156.25 or 156.25391 megahertz. This can be done either by our team or by the customer using our programmers. For large orders, we will efficiently program the devices in a high-volume facility and promptly ship them to the customer.

Programmability also enables optimisation for voltage, drive level, and other critical parameters in sys-

tem design. We introduced our first product in 2007. We have shipped about 35,000 unique part numbers worldwide across those 16 years, and every year it grows by 2000, 3000, and 4000 new part numbers.

Is the programmability factor the only difference between your MEMS technology and the traditional quartz?

I think the most significant difference is the resilience and reliability of the device—how resilient or robust it is to environmental disturbances like vibration, shock, temperature changes, airflow, etc. Our claim to fame in MEMS technology is that we are much more resilient and reliable than quartz devices. Of course, there is programmability, and the third one is that we’re a semiconductor company that uses the semiconductor supply chain, which is much more flexible than the quartz people who use dedicated factories. We manufacture at locations used by the semiconductor industry. For example, we use TSMC for our analogue, Bosch for our MEMS, and ASE for our packaging—all companies that